<u>Thm</u> (Riemann's rearrangement result) Let $\sum_{n=1}^{\infty} a_n$ be a series that converges, but does not converge absolutely. For any α, β satisfying $-\infty \le \alpha \le \beta \le \infty$, there exists a rearrangement $\sum_{n=1}^{\infty} \tilde{a}_n$ so that the partial sums $\tilde{s}_n = \sum_{k=1}^{n} \tilde{a}_k$ obey

$$\liminf_{n\to\infty} \tilde{s}_n = \alpha \quad \text{and} \quad \limsup_{n\to\infty} \tilde{s}_n = \beta.$$

<u>Pf</u>: (continued) Recall:

- $B_1, B_2, B_3, \ldots$ are the nonnegative terms of $\{a_n\}_{n\ge1}$
- $C_1, C_2, C_3, \ldots$ are the absolute value of the negative terms
- $\alpha_n \to \alpha$ as $n\to\infty$
- $\beta_n \to \beta$ as $n\to\infty$
- $\alpha_n < \beta_n \quad \forall n \ge 1$

Last time: We constructed a rearrangement

$$\begin{aligned}\sum_{n=1}^{\infty} \tilde{a}_n = {} & B_1 + \ldots + B_{k_1} - C_1 - \ldots - C_{j_1} \\ & + B_{k_1+1} + \ldots + B_{k_2} - C_{j_1+1} - \ldots - C_{j_2} \\ & + \ldots\end{aligned}$$

where at the nth step, we choose $k_n, j_n \in \mathbb{N}$ to be the smallest numbers s.t. the partial sums

$$x_n = B_1 + \ldots + B_{k_{n-1}+1} + \ldots + B_{k_n} > \beta_n$$

$$y_n = B_1 + \ldots \qquad \ldots + B_{k_n} - C_{j_{n-1}+1} - \ldots - C_{j_n} < \alpha_n$$

By construction we have

$$x_n - B_{k_n} \le \beta_n \quad \Rightarrow \quad \beta_n - B_{k_n} < \beta_n < x_n \le \beta_n + B_{k_n}$$

$$\Rightarrow \quad |x_n - \underbrace{\beta_n}_{\to \beta}| \le \underbrace{B_{k_n}}_{\text{subsequence of } \{a_n\}_{n\ge 1}} \to 0 \quad \text{as } n \to \infty$$

$\sum a_n$ conv. $\Rightarrow \lim_{n\to\infty} a_n = 0$

$$\Rightarrow \quad \lim_{n\to\infty} x_n = \beta$$

Similarly,

$$y_n + C_{j_n} \ge \alpha_n \quad \Rightarrow \quad \alpha_n - C_{j_n} \le y_n < \alpha_n < \alpha_n + C_{j_n}$$

$$\Rightarrow \quad |y_n - \underbrace{\alpha_n}_{\to \alpha}| \le C_{j_n} \to 0 \quad \text{as } n \to \infty$$

$$\Rightarrow \quad \lim_{n\to\infty} y_n = \alpha.$$

Finally, note that x_n and y_n are subsequences of the partial sums $\tilde{s}_n = \sum_{k=1}^{n} \tilde{a}_n$. So

$$\liminf_{n\to\infty} \tilde{s}_n \le \alpha \le \beta \le \limsup_{n\to\infty} \tilde{s}_n.$$

HW5#2 since $\{y_n\}$ is a subsequence of $\{\tilde{s}_n\}$

HW5#2 since $\{x_n\}$ is a subsequence of $\{\tilde{s}_n\}$

Exercise: $\liminf_{n\to\infty} \tilde{s}_n = \alpha$ and $\limsup_{n\to\infty} \tilde{s}_n = \beta$.

Idea: By construction, any intermediate partial sum is bounded by some x_n and y_n. E.g.,

$$\underbrace{B_1+\dots+B_{k_1}-C_1-\dots-C_{j_1}}_{y_1} \le \tilde{s}_n \le \underbrace{B_1+\dots+B_{k_1}}_{x_1} \quad \text{for } k_1 \le n \le k_1+j_1$$

$\square$

- So: in general, order matters!
- Here's a special case where it doesn't matter:

<u>Thm</u> If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then any rearrangement $\sum_{n=1}^{\infty} \tilde{a}_n$ converges to $\sum_{n=1}^{\infty} a_n$.

<u>Pf</u>: Set $s_n = \sum_{j=1}^{n} a_j$, $\tilde{s}_n = \sum_{j=1}^{n} \tilde{a}_j = \sum_{j=1}^{n} a_{k_j}$ for $n \ge 1$.

Fix $\varepsilon > 0$. As $\sum_{n=1}^{\infty} |a_n|$ converges, $\exists N_1 \in \mathbb{N}$ s.t.

$$\sum_{k=m}^{n} |a_k| < \frac{\varepsilon}{2} \quad \forall n \ge m \ge N_1 .$$

Set $N_2 = \max\{k_n : n = 1, 2, \dots, N_1\}$, so that

$$a_1, \dots, a_{N_1} \in \{\tilde{a}_1, \dots, \tilde{a}_{N_2}\}.$$

Then, for $n \ge N_2 + 1$ the terms $a_1, \dots, a_{N_1}$ are included in both s_n and $\tilde{s}_n$, and so

$$|s_n - \tilde{s}_n| \le \underbrace{\sum_{j=N_1+1}^{n} |a_j| + \sum_{\substack{j=1,\dots,n \\ \text{s.t. } k_j > N_1}} |a_{k_j}|}_{\text{all indices are } \ge N_1} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

As $\lim_{n\to\infty} s_n = s \in \mathbb{R}$, we conclude $\lim_{n\to\infty} \tilde{s}_n = s$ as well. $\square$

METRIC SPACES (Copson Ch. 2-3, Rudin Ch. 2)

- A metric is a notion of distance between 2 points

<u>Def</u> Let X be a nonempty set. A <u>metric</u> on X is a function $d: X \times X \to \mathbb{R}$ s.t.

① $d(x,y) \geq 0 \quad \forall x,y \in X$

② $d(x,y) = 0 \iff x = y$

③ $d(x,y) = d(y,x) \quad \forall x,y \in X$

④ (Triangle inequality) $d(x,z) \leq d(x,y) + d(y,z) \quad \forall x,y,z \in X$

Together, we call (X,d) a <u>metric space</u>.

<u>Ex</u> $X = \mathbb{R}$, $d(x,y) = |x-y|$ is a metric on $\mathbb{R}$.

<u>Ex</u> Let $X \neq \emptyset$ be any set. The discrete metric

$$d(x,y) = \begin{cases} 0 & x = y \\ 1 & x \neq y \end{cases}$$

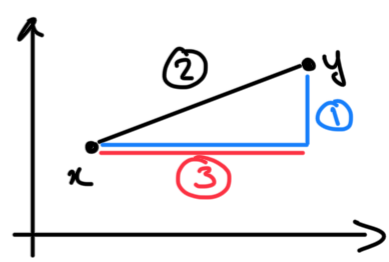
is a metric on X.

<u>Ex</u> $X = \mathbb{R}^2$, $x = (x_1, x_2)$, $y = (y_1, y_2)$ in $\mathbb{R}^2$.

① $d_1(x,y) = |x_1 - y_1| + |x_2 - y_2|$

② $d_2(x,y) = \sqrt{|x_1-y_1|^2 + |x_2-y_2|^2}$

③ $d_\infty(x,y) = \max\{|x_1-y_1|, |x_2-y_2|\}$



are all metrics on $\mathbb{R}^2$.

- Why do ①–③ satisfy the triangle inequality?
  Next time!